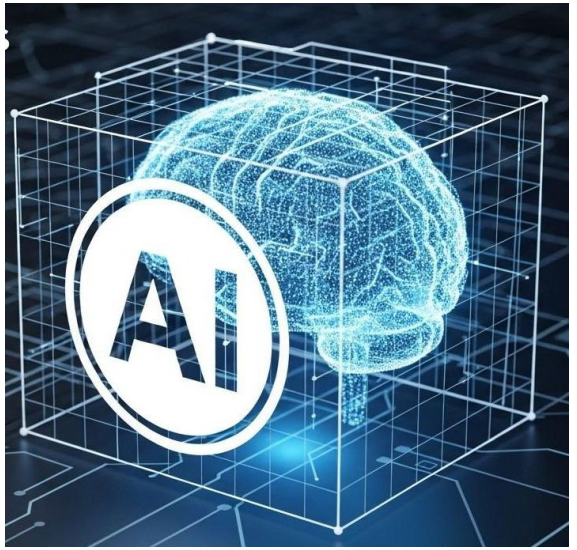
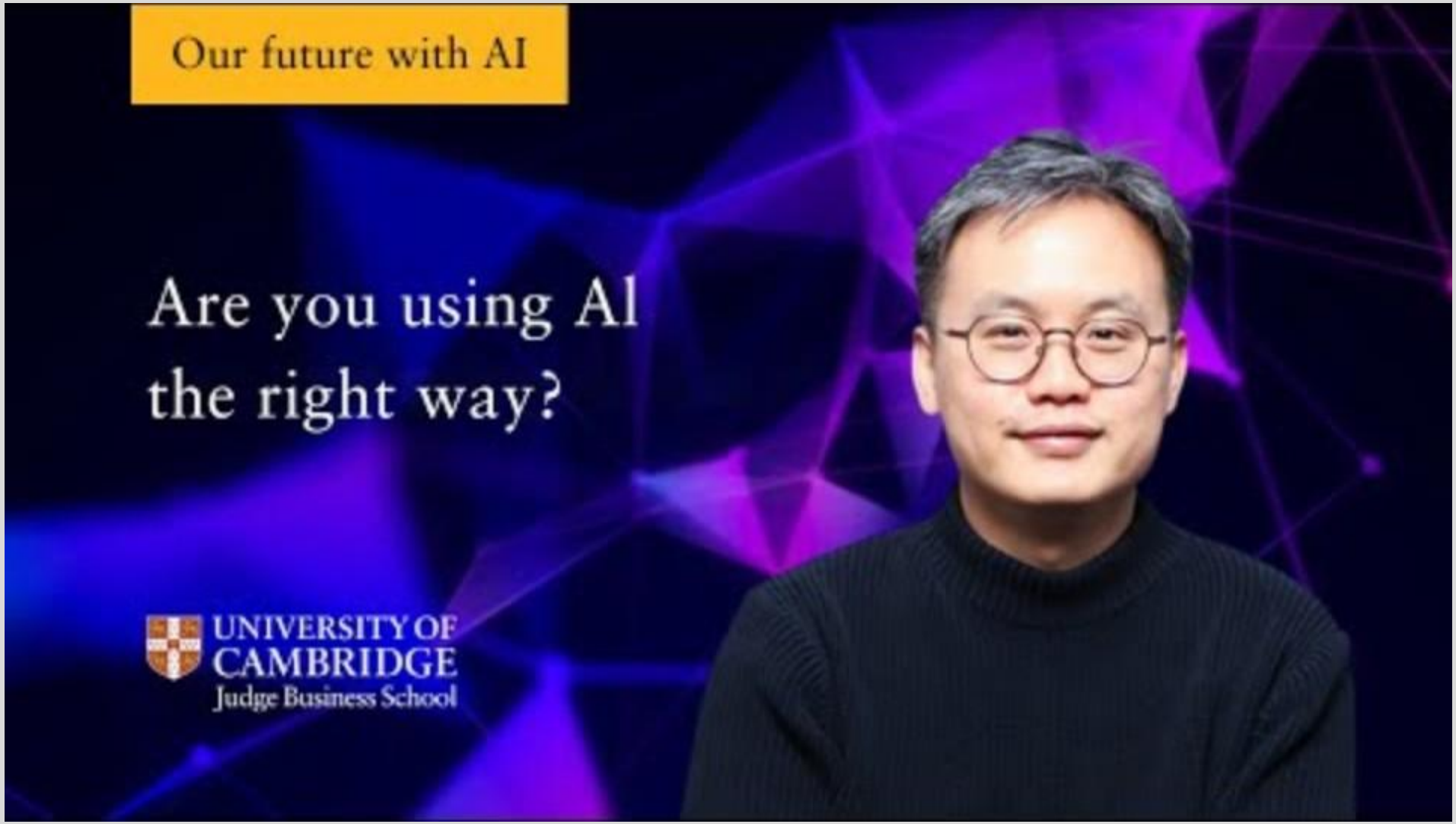


ALY 3510 · Fall 2026

Lecture 9: AI & Creativity

Can machines create — and what does that mean for us?

Pre-Lecture Video (2:05) - How to boost your creativity with generative AI



https://www.youtube.com/watch?v=QMzYgLcaLC4

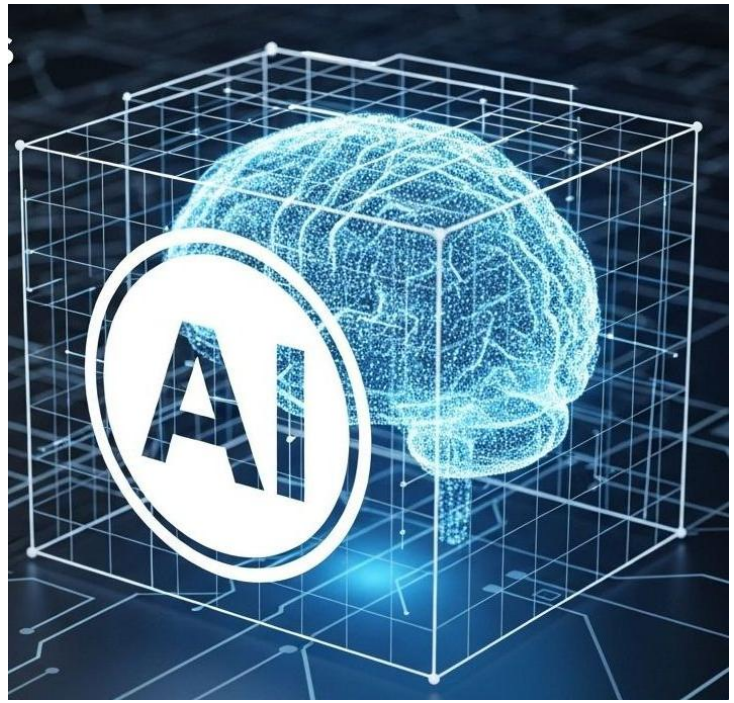
Agenda

Pre-lecture video

Lecture:

1. Foundations of Creative AI
2. AI in Visual Arts
3. AI in Music Composition
4. AI in Creative Writing
5. Human-AI Collaboration
6. AI and Scientific Creativity
7. Summary & Action Steps

In-class discussion





Part 1: Foundations of Creative AI

Defining creativity and how machines approach it

https://datasciencedojo.com/newsletter/ai-generated-music/

What is Creativity?

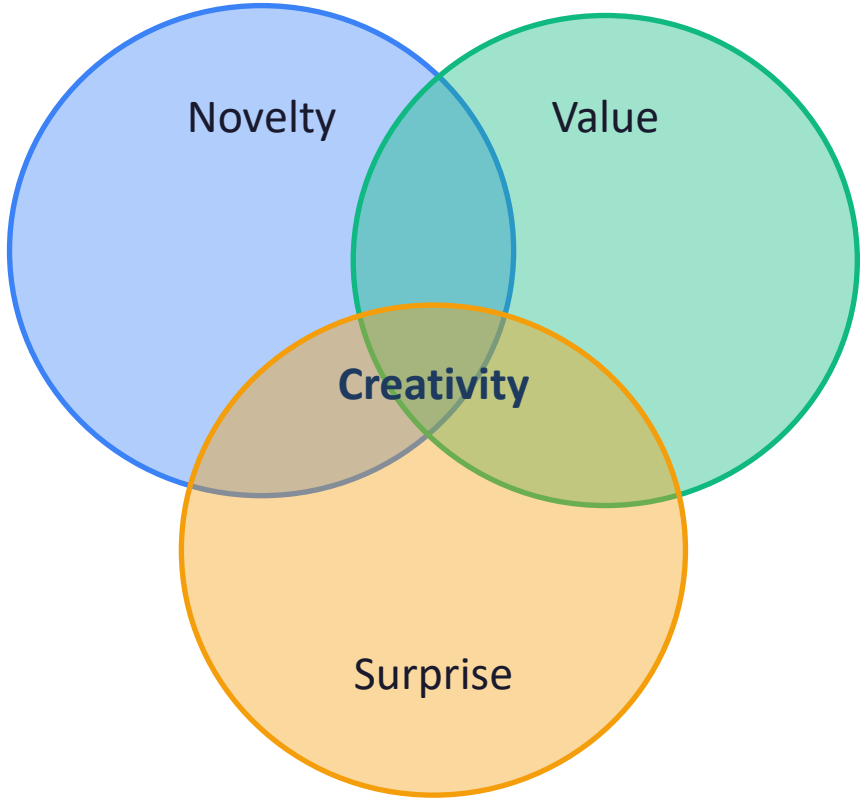
Boden's Criteria

Creativity is producing ideas or artifacts that are:

Novel
New and original

Surprising
Unexpected combinations

Valuable
Meaningful or useful



Boden, M. (2004) - The Creative Mind

The central question of this lecture: if an AI produces something novel, valuable, and surprising — is that creativity, or just sophisticated pattern-recombination?

Can Machines Be Creative?

"The Analytical Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform." — The Lovelace Objection (Ada Lovelace, 1843)

The skeptical view

- Real creation requires understanding & intention
- AI recombines patterns from its training data
- Novelty is not the same as originality

The modern counterpoint

- Computational creativity: programs that build, understand, and enhance creative work
- Outputs can genuinely surprise their creators
- Is combining patterns differently ‘true creation’?

Where AI Helps Most: The Co-Pilot Model

AI as co-pilot

- Humans ideate first; AI refines & generates variations
- Text-to-image AI enhances human creative productivity by 25%
- "We find that generative AI significantly boosts artists' productivity and leads to more favorable evaluations from their peers."

Research insights

- Peak novelty rises as artists master AI tools
- But average novelty narrows — risk of homogenization
- Human-first ideation preserves creative diversity

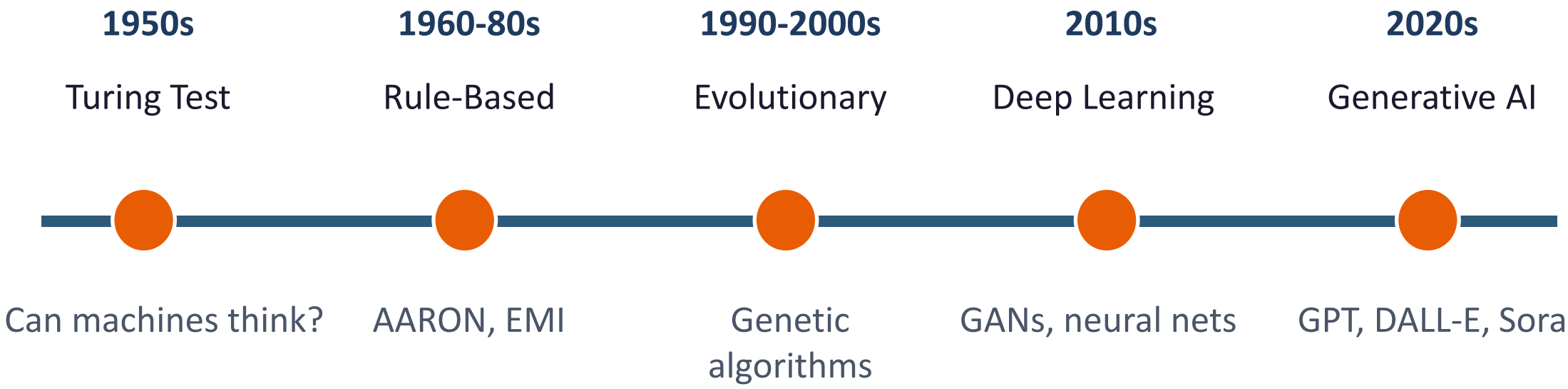
Best practice: use AI as a collaborator, not a replacement — human judgment guides the creative direction.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4594824

academic.oup.com (Zhou & Lee, 2024)

Evolution of Creative AI

From Turing's question to today's generative tools:



Key Shift: From rule-following to pattern-learning to genuine generation

How AI Creates: Three Key Technologies

GANs

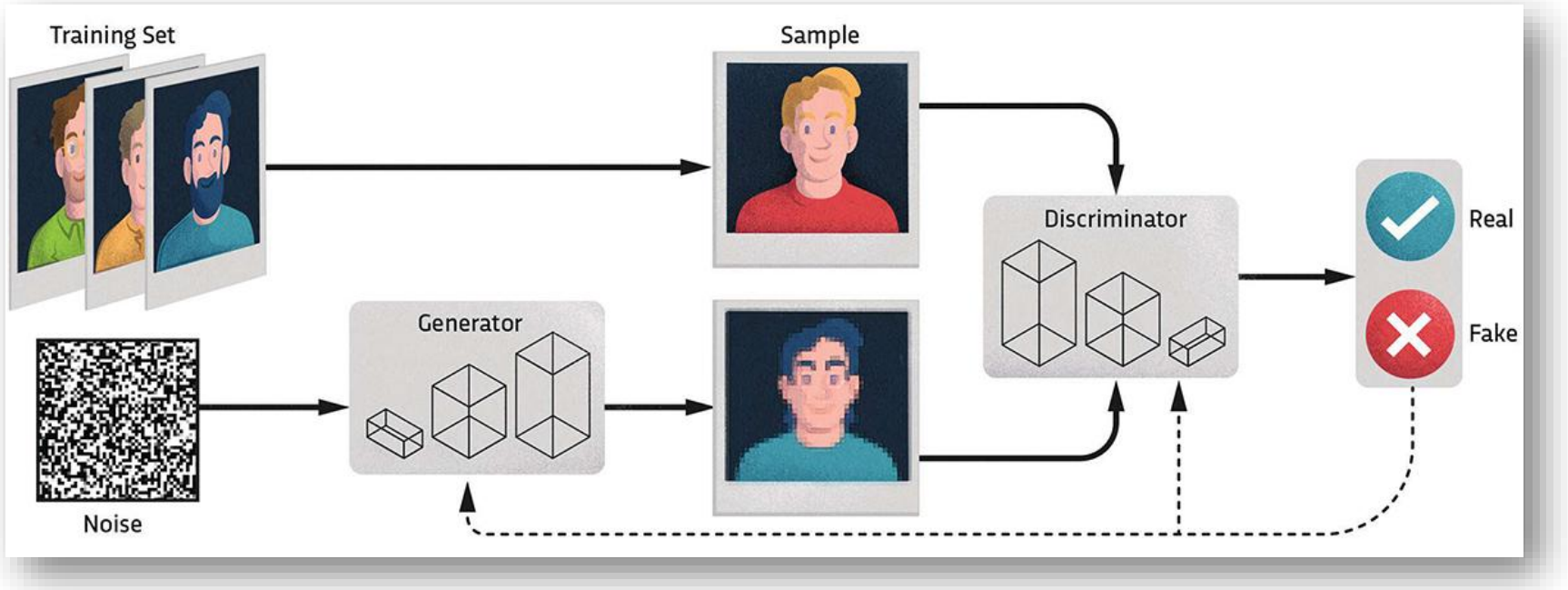
Two networks compete — one creates, one critiques — until outputs look real.

Transformers

Learn patterns in sequences — generate coherent text, music, and code.

Diffusion

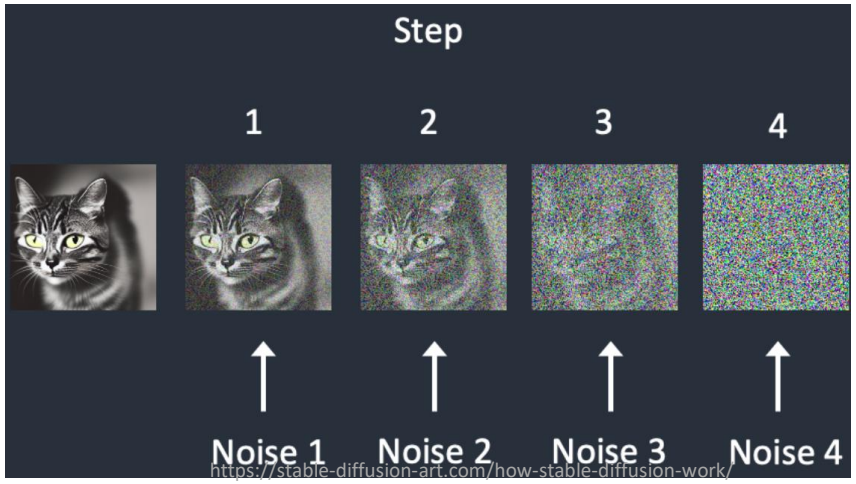
Learn to reverse noise — turn random noise into images. Powers modern image tools.

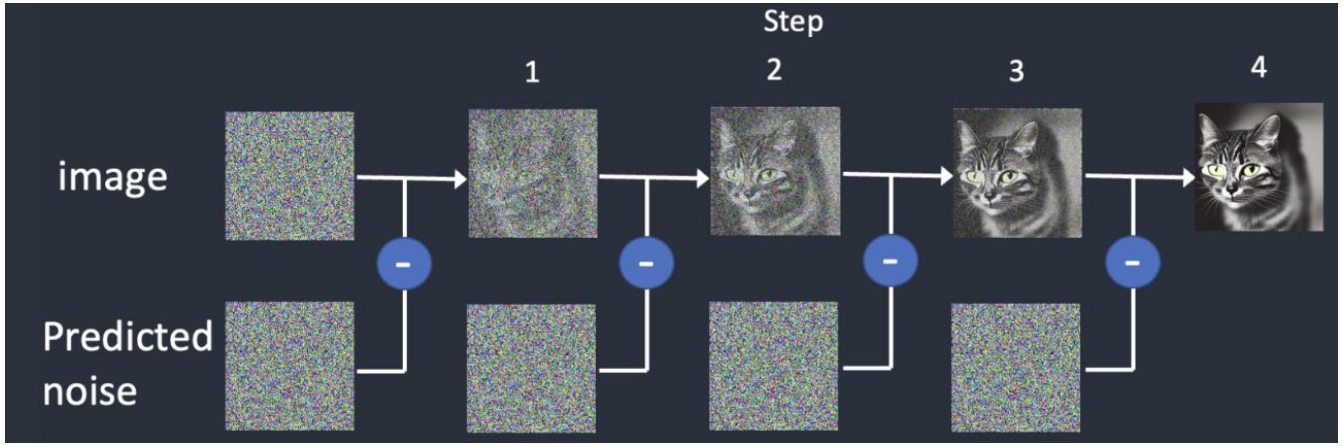


Each approach enables different creative applications — often combined in modern systems.

How Stable Diffusion Works

- Forward process: noise is added to an image, step by step, until it's pure noise — this trains a "noise predictor."
- Reverse process (generation): start from random noise, predict & subtract the noise, guided by your text prompt — a new image emerges.
- **The key insight: by learning to predict noise, the model learns the structure of images.**





Reverse diffusion: denoise toward the prompted image.

How Generative AI Works (5:17)

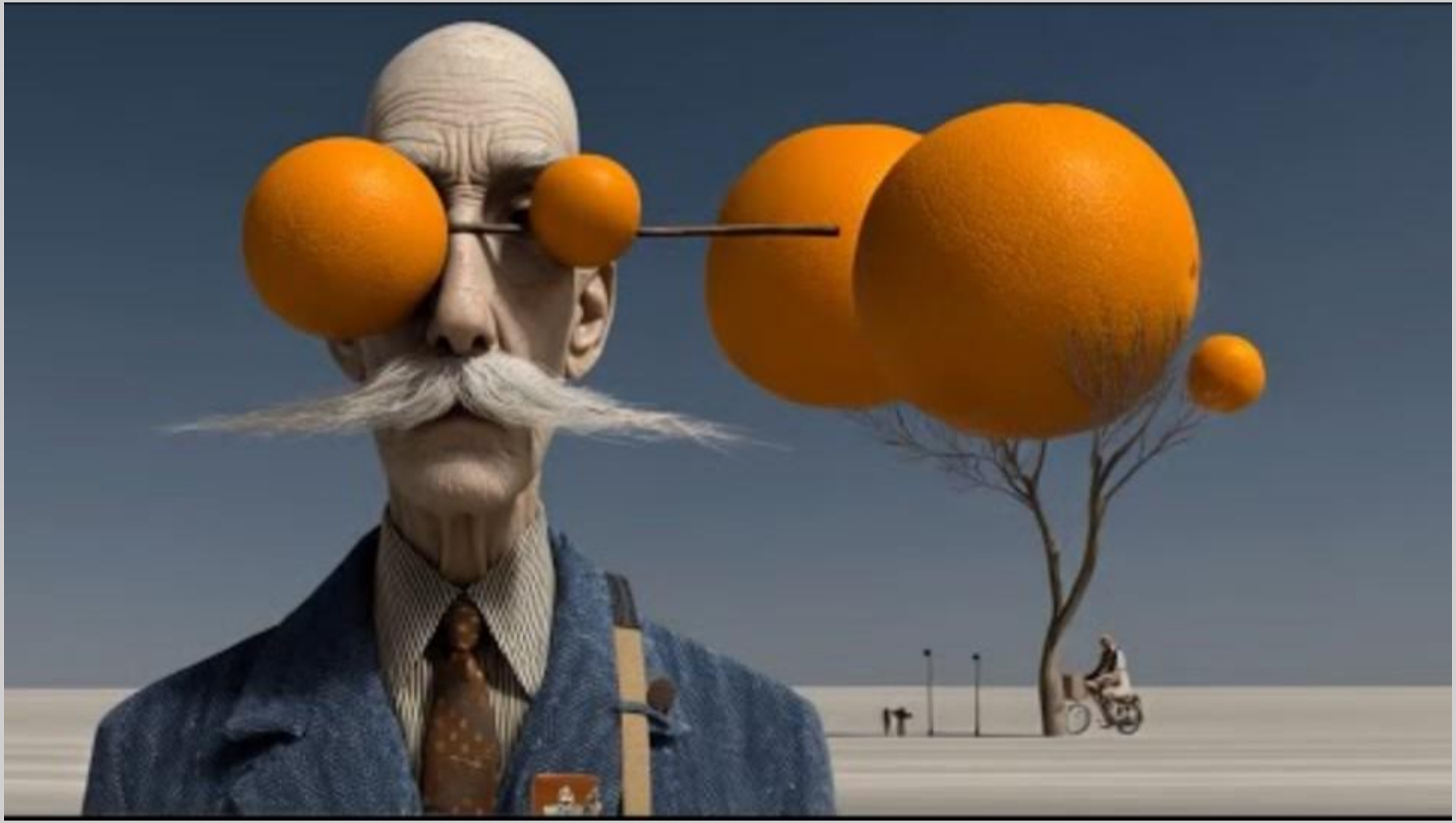


https://www.youtube.com/watch?v=X994dDnmRmY

Part 2: AI in Visual Arts

The revolution in digital art creation

AI in Visual Arts Video (2:05)



As you watch: What makes this art "creative"?

https://www.youtube.com/watch?v=xJSTIwDlBPM&list=RDhVi_qpxX6YQ&index=2

AI in Visual Arts: The Revolution

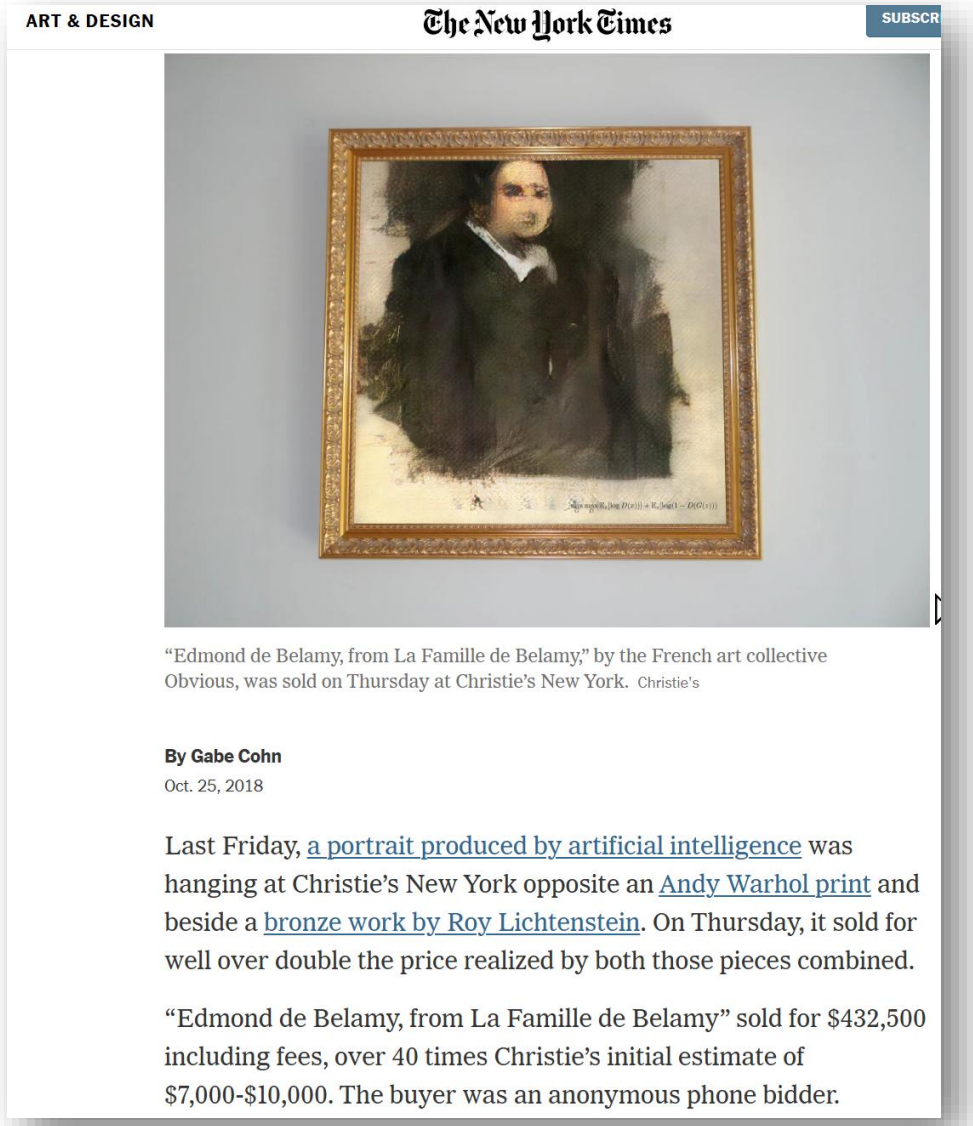
Same prompt, different outputs—each platform has a distinct style

DALL·E 3 (OpenAI)	Midjourney	Stable Diffusion	Adobe Firefly
			
Photorealistic; replaced by ChatGPT Images	Highly stylized, artistic outputs	Open-source experimentation	Trained on licensed content

Tools change fast: OpenAI shipped three ChatGPT Images versions in 2026 alone.

AI Art is Selling — and the Novelty Paradox

- AI-authored artwork now commands real money — “Edmond de Belamy” sold for $432,500 at Christie’s (2018).
- **The novelty paradox: the best AI art is more creative than ever — but most AI art looks the same.**
- Creative space is expanding, but with inefficiencies and a risk of homogenized content.
- Success requires strong ideation and filtering skills — human judgment still matters.



ART & DESIGN

The New York Times

SUBSCR

“Edmond de Belamy, from La Famille de Belamy,” by the French art collective Obvious, was sold on Thursday at Christie’s New York. Christie’s

By Gabe Cohn

Oct. 25, 2018

Last Friday, a portrait produced by artificial intelligence was hanging at Christie’s New York opposite an Andy Warhol print and beside a bronze work by Roy Lichtenstein. On Thursday, it sold for well over double the price realized by both those pieces combined.

“Edmond de Belamy, from La Famille de Belamy” sold for $432,500 including fees, over 40 times Christie’s initial estimate of $7,000-$10,000. The buyer was an anonymous phone bidder.

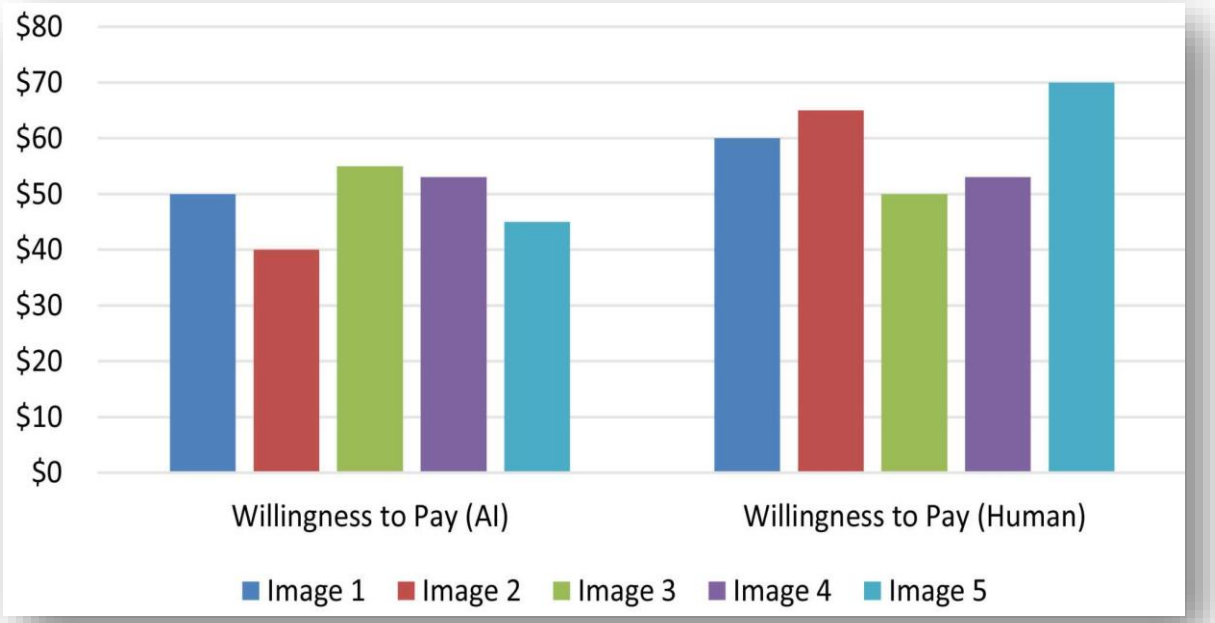
https://www.nytimes.com/2018/10/25/arts/design/ai-art-sold-christies.html

Perception & Value of AI Art

Authenticity matters more than novelty in determining perceived artistic value.

- AI art is seen as more novel but less authentic.
- The greatest stigma is when AI is used for execution; less so when used only for ideation.
- Artists using AI can be perceived as less skilled — and people will pay less.
- Commercial art is judged less harshly than “high art.”

Willingness to Pay Chart



https://www.scirp.org/pdf/adr2025131_51250604.pdf

AI in Film and Animation (2:53)

Key Applications

- Visual Effects
- Storyboarding
- Voice Synthesis
- Video Generation

Platforms: Runway ML, Veo, Sc



Example: "Watch the Skies" (Sweden) used AI to sync lip movements with dubbed English dialogue

https://www.youtube.com/watch?v=PTngv5MmtXo

Part 3: AI in Music Composition

Composing the future of sound

AI Music: Tools & Uses

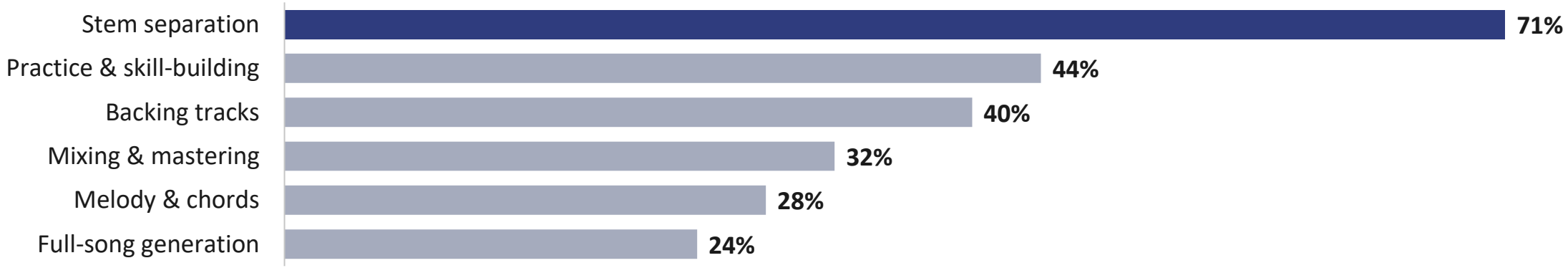
Licensing in 2026

- Universal settled with Udio (Oct 2025)
- Warner settled with Suno and Udio (Nov 2025)
- Udio became a closed app: no downloads
- Sony is still in court against both

Generation tools

- Suno — full songs with vocals (v5.5, Mar 2026)
- Udio — remix licensed music inside the app
- AIVA — soundtracks for film & games
- ElevenLabs — commercially cleared songs

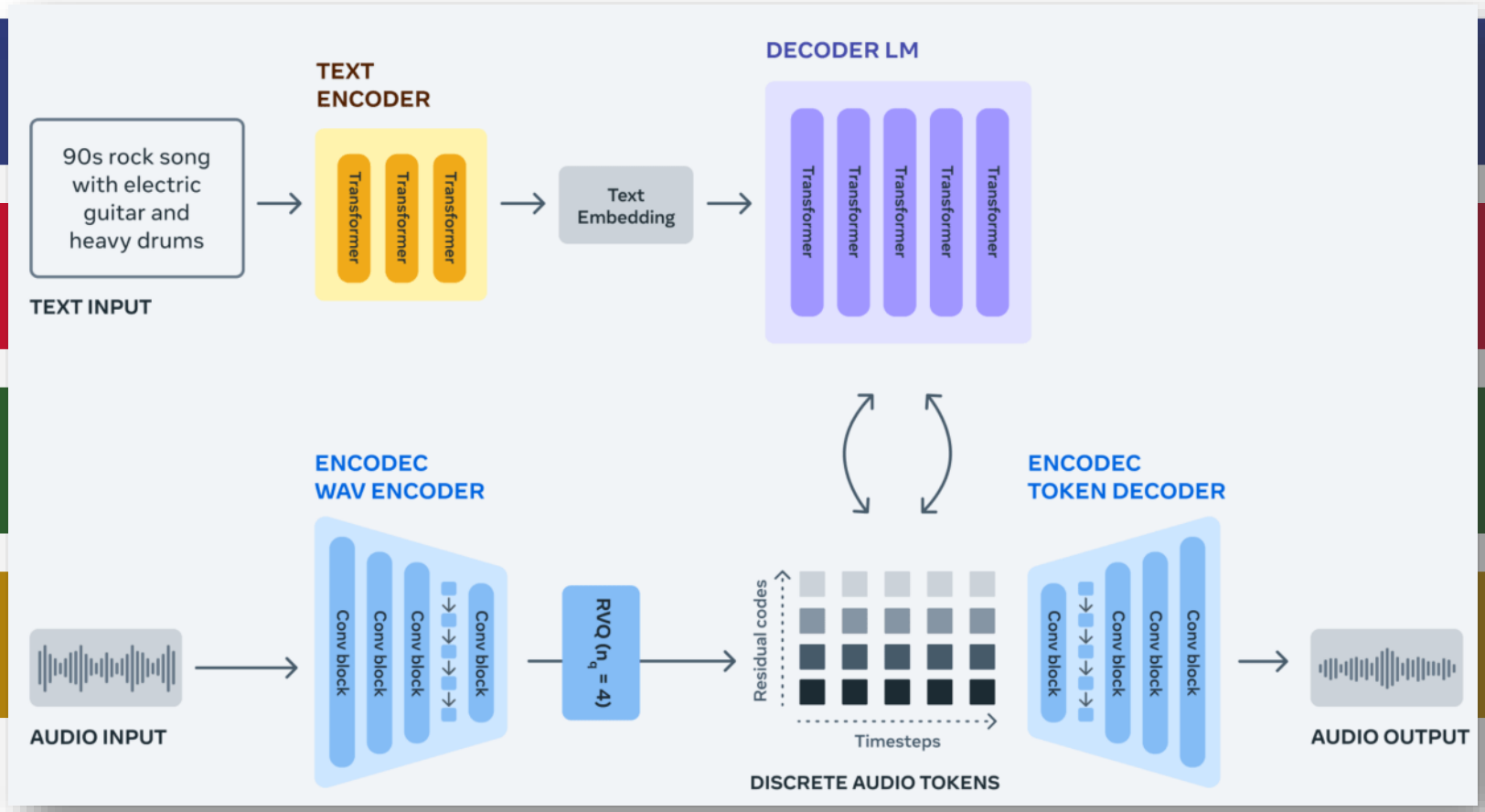
What musicians use AI for, share of AI users, late 2025



Source: Water & Music for Moises, “Tools, not tricks” survey of 1,525 musicians (fielded late 2025, published Mar 2026); shares are among the 1,021 who use AI.

How AI Music Generation Works

MusicGen (Meta) architecture — in plain English:



Source: datasciencedojo.com/newsletter/ai-generated-music/

AI Music Generation (0:31)



https://www.youtube.com/watch?v=v6q4MjUB0Qo

Controversies and Concerns

The Luxembou

When AIVA-com
called it

Ongoing Debates

- Does AI music lac
- Will AI eliminate
- Can AI replicate a
- Who owns AI-gen



https://iveinnovation.org/aiva-technology-extraordinary-ai-music-start-up/

https://www.youtube.com/watch?v=H6Z2n7BhMPY

Part 4: AI in Creative Writing

From poetry to screenplays

AI as a Writing Partner

Ideation

Brainstorm plots, themes, story arcs — directions you might not consider.

Character development

Backstories, motivations, dialogue; probe fears & desires.

World-building

Settings, cultures, rules — and their implications.

Tools: Sudowrite, GravityWrite, Novelcrafter, ChatGPT/Claude — for plot outlines, meter & rhyme, style mimicry, and overcoming creative blocks.

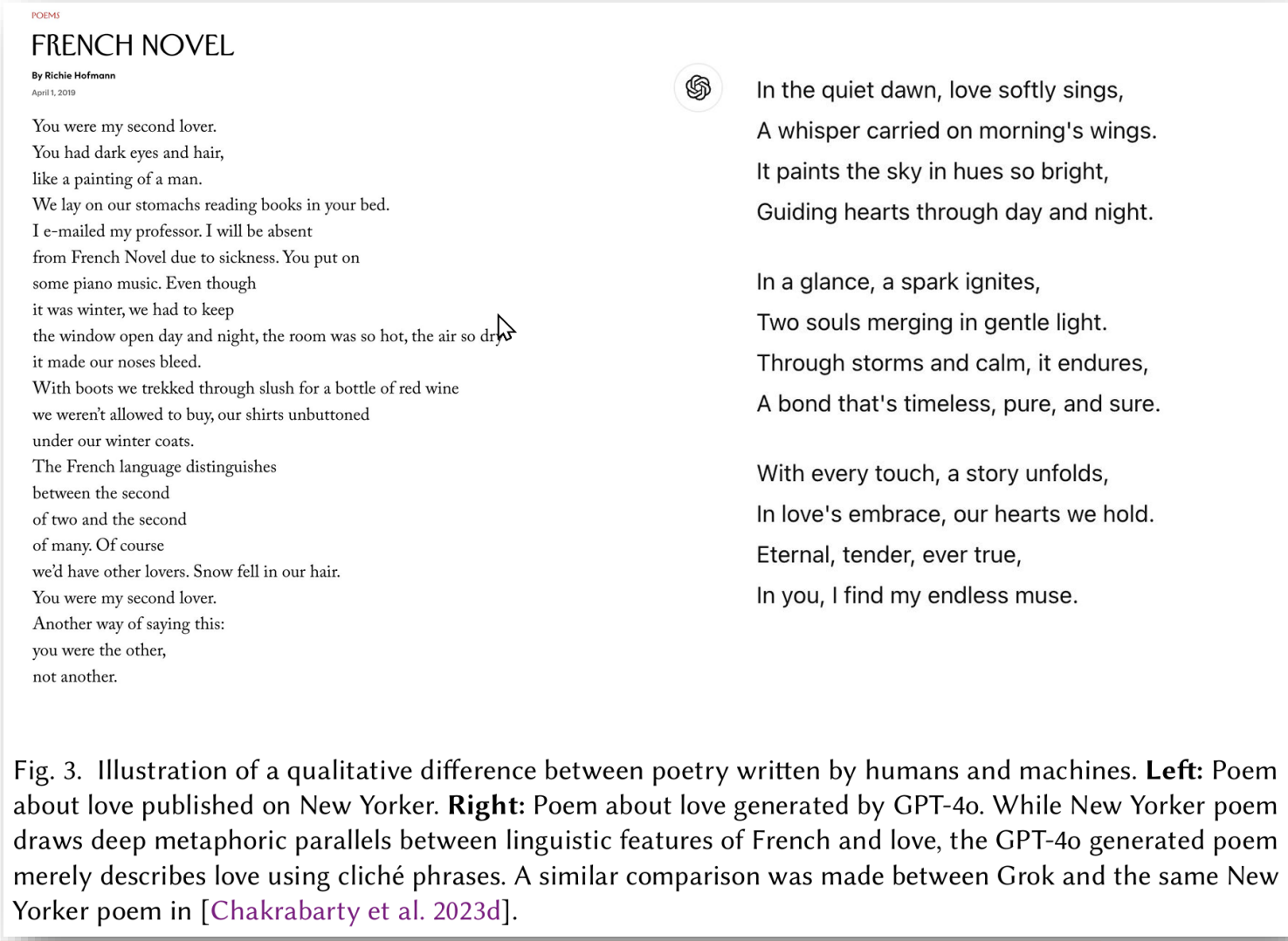


Practice: use AI for exploration and iteration — your vision and voice guide the creative direction.

https://tinyurl.com/2ye29yl7

Poetry: Human vs. AI

- A study compared poems written by humans vs. GPT-generated poems in the New Yorker style.
- Human poems drew deep metaphorical parallels; AI poems more often described feelings with cliché phrases.
- **There is often a qualitative difference — but it's getting harder to spot.**



POEMS

FRENCH NOVEL

By Richie Hofmann

April 1, 2019

You were my second lover.
You had dark eyes and hair,
like a painting of a man.
We lay on our stomachs reading books in your bed.
I e-mailed my professor. I will be absent
from French Novel due to sickness. You put on
some piano music. Even though
it was winter, we had to keep
the window open day and night, the room was so hot, the air so dry
it made our noses bleed.
With boots we trekked through slush for a bottle of red wine
we weren't allowed to buy, our shirts unbuttoned
under our winter coats.
The French language distinguishes
between the second
of two and the second
of many. Of course
we'd have other lovers. Snow fell in our hair.
You were my second lover.
Another way of saying this:
you were the other,
not another.

In the quiet dawn, love softly sings,
A whisper carried on morning's wings.
It paints the sky in hues so bright,
Guiding hearts through day and night.

In a glance, a spark ignites,
Two souls merging in gentle light.
Through storms and calm, it endures,
A bond that's timeless, pure, and sure.

With every touch, a story unfolds,
In love's embrace, our hearts we hold.
Eternal, tender, ever true,
In you, I find my endless muse.

Fig. 3. Illustration of a qualitative difference between poetry written by humans and machines. **Left:** Poem about love published on New Yorker. **Right:** Poem about love generated by GPT-4o. While New Yorker poem draws deep metaphoric parallels between linguistic features of French and love, the GPT-4o generated poem merely describes love using cliché phrases. A similar comparison was made between Grok and the same New Yorker poem in [Chakrabarty et al. 2023d].

Discussion: Which would you rather read? Can you tell the difference? Why does it matter?

https://arxiv.org/pdf/2410.17218

Narrative Diversity Risks

The homogenization problem: The risk is huge datasets dominated by majority content.

Key risks

- Overrepresentation of dominant cultures
- Loss of unique regional voices
- Repetitive narrative structures
- Bias toward ‘popular’ storylines

Writer’s responsibility

- Actively curate prompts for diversity
- Question AI suggestions critically
- Inject your unique perspective
- Research beyond AI’s training data

Can AI write a screenplay? (0:56)



https://www.youtube.com/shorts/ft3zhDl8f-M

BREAK: Image To Video AI Free Tool (1:05)



https://www.youtube.com/watch?v=zNV_89TVenU

Part 5: Human-AI Collaboration

Partners, not replacements

The Partnership Model

AI serves as a creative partner, not a replacement — co-creativity in which each contributes unique strengths.

AI's role	What it contributes
Ideation assistant	Suggests new directions
Skill amplifier	Work beyond your technical limits
Efficiency tool	Automates repetitive tasks
Exploration engine	Generates variations to refine



Tools for everyone

- No coding required — natural-language prompts
- Templates & real-time feedback
- Democratizes professional-quality output
- **Before AI: years of training. With AI: minutes to create.**

AI-powered Creativity (2:44)



https://www.youtube.com/watch?v=8eWXkCyHce8

Ethical Concerns: Copyright and Ownership

Key questions

- **Training data:** is training on copyrighted work without permission fair use?
- **Authorship:** how much human input makes AI-assisted work copyrightable?
- **Outputs:** who is liable when an AI output copies a protected work?
- **Consent:** should artists be able to opt out of training sets?

U.S. legal status, 2026

- **No AI authors:** Supreme Court declined Thaler's appeal (Mar 2026)
- **Prompts alone** aren't authorship; human edits and arrangement can be protected
- **Fair use:** training on lawfully bought books (Bartz v. Anthropic, 2025)
- **Not fair use:** training on Westlaw summaries (Thomson Reuters v. ROSS, 2026)

Still unsettled: courts disagree on training data, and Congress has passed no AI copyright law.

Sources: U.S. Copyright Office, Copyright and Artificial Intelligence, Part 2: Copyrightability (Jan 2025); Thaler v. Perlmutter, cert. denied (Mar 2, 2026); Bartz v. Anthropic (N.D. Cal., Jun 2025); Thomson Reuters v. ROSS Intelligence (3d Cir., Sep 2026).

Ethical Concerns: Artist Displacement & Bias

The displacement debate

- Jobs: AI replacing entry-level creative roles
- Style mimicry: replicating artists' styles without consent
- 6,500+ artists petitioned against "mass theft" of training data

Bias in AI creativity

- Training data reflects historical prejudices
- Mainstream aesthetics get amplified
- E.g., "librarian" prompts skew to older white women

Historical context: every creative technology (photography, synthesizers, digital tools) faced similar debates — but scale and consent make this one different.

Protecting Human Creativity

Tool / practice	What it does
Glaze	Invisibly alters artwork to prevent AI mimicry
Nightshade	“Poisons” AI training data to disrupt mimicry
“Do Not Train” + provenance	Content Credentials (C2PA) to mark and verify human work

Best practices for ethical AI creativity

Transparency (disclose AI use) · attribution & consent · augmentation · critical evaluation.

Adobe Firefly trains only on licensed and public domain content — an ethical-by-design example.



https://cybernews.com/ai-news/glaze-nightshade-cant-stop-ai-scraping-art/

Success Stories of AI Collaboration (1:00)

Notable Collaborations

- **Holly Herndon** — AI-created vocal techniques
- **VANDAL "Natural Rhythms"** — 130-min AI documentary
- **"Beautiful the World"** — Won the 2020 AI Song Contest
- **Christie's AI Sale** — First dedicated AI auction



https://vimeo.com/886680053

These examples show AI's potential when used thoughtfully in creative projects.

Part 6: AI and Scientific Creativity

From proteins to theorems

Scientific Creativity & AlphaFold

- Scientific creativity: generating novel ideas, approaches, or solutions that lead to new discoveries.
- AlphaFold (DeepMind) solved a 50-year problem — predicting protein structure — in months, not years.
- 200M+ protein structures predicted, transforming biology.



https://tinyurl.com/26doe85l

2024 Nobel Prize in Chemistry: awarded to Demis Hassabis & John Jumper (AlphaFold) and David Baker (protein design) — the second Nobel recognizing an AI system's scientific contribution. The Physics prize to Hopfield and Hinton was announced the day before.

AI in Drug Discovery

The traditional challenge

- 10–15-year timelines
- Up to $2.8 billion in cost
- 80–90% failure rate

How AI helps

- Target identification
- Molecular & property design
- Synthesis planning

Success story: Insilico Medicine used AI to discover a novel target and design a molecule for pulmonary fibrosis in ~30 months — reaching Phase II trials. Platforms: Insilico, Exscientia (Recursion), Isomorphic Labs, BenevolentAI.

The Centaur Approach: Human-AI Partnership

"We want to encode drug discovery, but we need to understand human expertise and tacit knowledge, and ask how that becomes something codable, repeatable, and scalable." — Andrew Hopkins, Exscientia

Humans provide

Scientific intuition, contextual understanding, hypothesis formation.

AI provides

Computational power, pattern recognition, vast search.

Together

Breakthroughs are easier when together.



https://en.wikipedia.org/wiki/Advanced_chess



https://tinyurl.com/285npenp

AI in Math & Physics

- Automated theorem provers generate proofs, check validity, and solve competition-level problems.
- AI helped discover new relationships in knot theory and suggested a path to a 40-year-old conjecture — a significant AI contribution to pure math research (2021).
- **2025: DeepMind and OpenAI systems reached gold-medal performance at the International Mathematical Olympiad.**



Each step of a proof can be checked by a machine.

AI in Math & Physics: Update (Late 2026)

OpenAI

- **Navier–Stokes blowup proof** from a swarm of **~10,000 agents** in 88 hours; Lean version disputed (Sep 8)
- Released **722 manuscripts** on 300+ open problems in one GitHub drop (Oct 6)
- Claims a proof of Sogge's **local smoothing conjecture** in every dimension, which also settles Kakeya (Oct 26)
- Claims an **existence proof for 4D QCD**, now under expert review

Anthropic

- **Dozens of Claude agents** built the first **machine-checked Fermat's Last Theorem proof** in 11 days (Sep 4)
- Claude computed a **nine-loop particle-scattering amplitude**; the record was eight (Sep 25)

Where each result stands

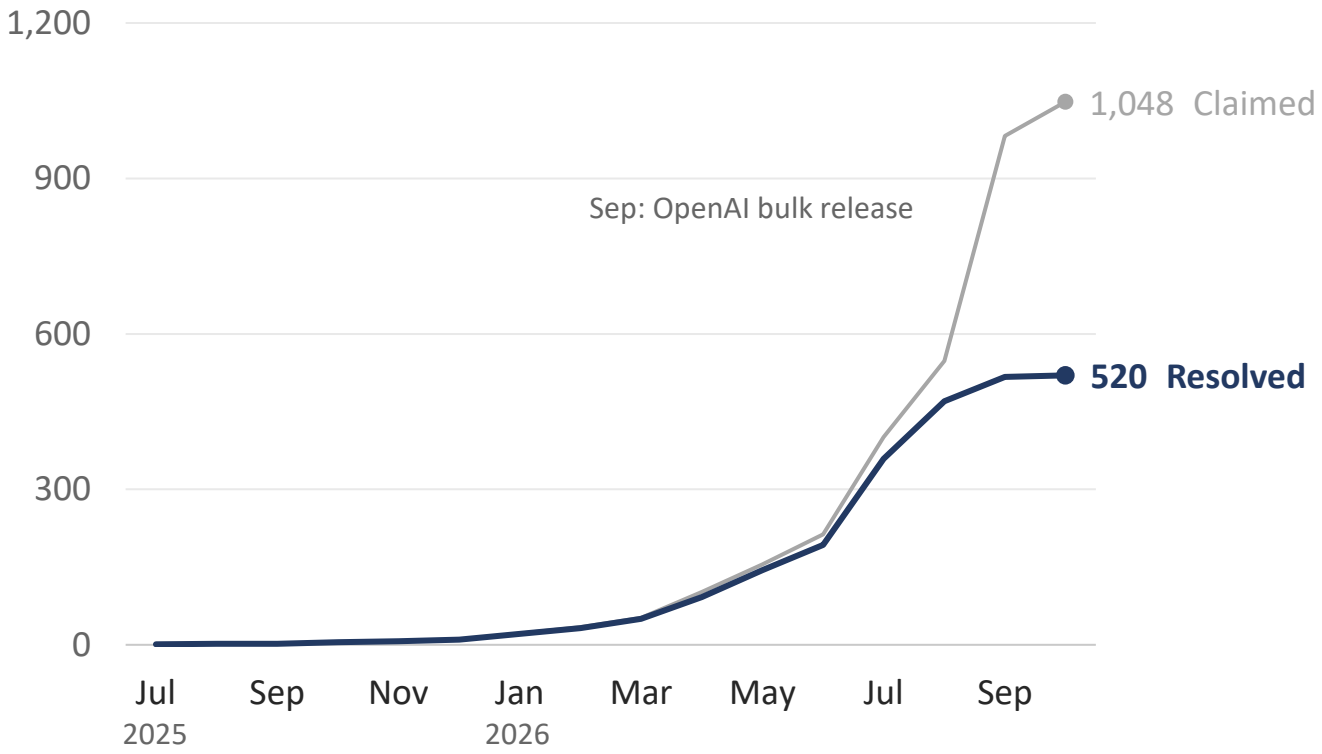
	Formal proof	Expert check	Next step
Fermat's Last Theorem Anthropic	Done	Done	Adds 29,500 formal theorems
Nine-loop amplitude Anthropic	n/a	Done	Test on harder amplitudes
Navier–Stokes blowup OpenAI	Disputed	Pending	Match Lean to written proof
Kakeya conjecture OpenAI	Done	Pending	OpenAI-funded review conference
4D QCD existence OpenAI	Withheld	Pending	Physicists' write-up

Sources: OpenAI announcements (Sep 8, Oct 6 and Oct 26, 2026); Anthropic research posts (Sep 4 and Sep 25, 2026); King's College London preprint on the Navier–Stokes Lean proof (Oct 2026).

What's next: expert review decides which claims stand. OpenAI withdrew three manuscripts on Oct 7 after a sign error, so treat unreviewed results as provisional.

AI went from 10 to 520 solved open math problems in 2026

Open math problems solved with AI in the loop, cumulative count



- Solutions took off in **mid-2026**: 166 problems resolved in July alone
- Claims now outpace checking: **528 more** await independent review
- OpenAI models are named on **425** resolved entries, Claude on **111**, Google DeepMind on 36

Note: counted by solve date; “claimed” adds candidate solutions not yet checked by anyone independent. Source: VibeMathed community dataset (vibemathed.com/stats), accessed Nov 7, 2026.

What to watch: whether the resolved line catches up with the claims as reviewers work through the 528-claim backlog.

Part 7: Summary & Action Steps

Your creative-AI journey starts now

The Future of AI Creativity Tools

Multimodal generation

Now: Veo 3 and Sora 2 make video with synced dialogue and sound effects

Next: multi-shot scenes you can edit one shot at a time

Real-time collaboration

Now: Google's Lyria RealTime "jams" music live, reacting within about 2 seconds

Next: AI bandmates and co-editors in live shows

Personalized style

Now: Suno v5.5 adds custom vocal personas; ChatGPT Images keeps characters consistent across edits

Next: rules on consent for whose voice and style a model learns

Licensed models

Now: Adobe Firefly trains on licensed content; Universal and Warner licensed Udio

Next: Label-licensed Suno and Udio models, promised for 2026

Discussion: which of these would you use in your own work, and which would you refuse?

Sources: Google DeepMind (Veo 3, May 2025; Lyria RealTime, 2025); OpenAI (Sora 2, Sep 2025; ChatGPT Images 2.6, Nov 2026); Suno v5.5 release (Mar 2026); Music Business Worldwide on label deals (Apr 2026).

Action Steps: Engaging with AI Creativity

For Students and Professionals

1	**Experiment**	Make one piece with a free tool
2	**Reflect**	Ask whose work trained the model
3	**Collaborate**	Add your own edits and choices
4	**Advocate**	Disclose when you use AI
5	**Learn**	Follow copyright and licensing news

Starting Points

Free tools to explore:

- ChatGPT Images (images)
- Gemini with Lyria 3 (songs)
- MusicFX DJ (live music)
- Claude (writing)
- Meta AI (Midjourney images)

Check age limits and terms first

Conclusion: The Creative Future

AI transforms creative expression A powerful tool across visual arts, music, writing, and science.

Ethical questions matter Authorship, ownership, and human value need ongoing discussion.

Collaboration over replacement Success lies in a thoughtful human-AI partnership.

We shape the future Active participation decides how AI integrates into creative culture.

The future of creativity will be defined not by AI's capabilities alone, but by how we choose to use, regulate, and integrate these tools — while preserving what makes human creativity valuable.

Let's Discuss (5–10 min)

1. If an AI produces a poem or painting that moves you, and you can't tell it was AI-made — is it "creative"? Does knowing a machine made it change its value to you? Why?

2. Artists whose work trained these models weren't asked or paid. Is training on public art "learning" (like a human studying masters) or "theft"? Where should the line be — and who should draw it?

Alternate: AlphaFold won a Nobel for a scientific discovery. Is that a different kind of "creativity" than making art — or the same thing?